The Mathematics of Money

Chapter 1: Simple vs. Compound Interest and the Time Value of Money

Study Notes Series

1 A Dollar Today Is Worth More Than a Dollar Tomorrow

The single most important idea in personal finance is the **time value of money**: money available now can be invested and grow, so it is worth more than the same amount received later. Interest rates are the price attached to this trade-off between present and future money.

2 Simple Interest

Simple interest grows linearly: interest is earned only on the original principal, never on previously earned interest.

$$A = P(1 + rt) \tag{1}$$

where P is the principal, r is the annual interest rate (as a decimal), t is time in years, and A is the final amount.

3 Compound Interest

Compound interest earns "interest on interest." If interest is compounded n times per year:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}. \tag{2}$$

As the compounding frequency n increases without bound (compounding continuously), this expression converges to:

$$A = Pe^{rt}. \tag{3}$$

This is the same exponential constant $e \approx 2.71828$ that appears throughout mathematics and physics; compound growth and exponential growth are the same phenomenon.

Why the difference matters over long horizons

Consider $P = \$1000$ invested at $r = 5\%$ for $t = 30$ years.

- Simple interest: $A = 1000(1 + 0.05 \times 30) = 1000(2.5) = \$2500$.

- Compound interest (annual compounding, $n = 1$): $A = 1000(1.05)^{30} \approx \$4321.94$.

Over 30 years, compounding nearly doubles the outcome compared to simple interest, even though the rate is identical. The gap widens dramatically over longer horizons, which is why starting to invest early matters more than the exact rate of return.

4 The Rule of 72

A handy mental shortcut estimates how long it takes an investment to double under compound interest:

$$t_{\text{double}} \approx \frac{72}{R} \tag{4}$$

where R is the interest rate expressed as a whole number (e.g., $R = 6$ for 6%). At 6% annual growth, money doubles roughly every $72/6 = 12$ years. This approximation comes from the fact that $\ln(2) \approx 0.693$, and 72 is chosen for its divisibility rather than mathematical precision.

Quick check

At an annual compound rate of $r = 8\%$, roughly how many years does it take an investment to double? (Answer: using the rule of 72, $t \approx 72/8 = 9$ years.)